Magnetism in Rocks – Background Information and Additional Resources

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Background information:

- The majority (94%) of Earth's magnetic field is generated from movement of liquid iron in the outer core. However, a small amount (around 3%) comes from magnetised rocks in the lithosphere (Olsen & Pauluhn, 2019).

- **Crustal (rock) magnetism** has 2 parts:
  1. *Induced magnetism* – active magnetism induced in rocks exposed to Earth's present magnetic field.
  2. *Remanent magnetism* – fossil magnetism preserved in rocks as they formed.

  (Olsen & Pauluhn, 2019)

- **Remanent magnetism** is the key concept in this class. Remanent magnetism is created when iron-rich magnetic minerals align with Earth's magnetic field during rock formation (Olsen & Pauluhn, 2019).

- The strength of a rock's magnetism depends on the amount of magnetic minerals it contains. The most common magnetic mineral is **magnetite.** Other magnetic minerals include metallic iron, nickel, cobalt, magnetite, pyrrhotite, and ulvospinel (University of British Columbia, 2018).

- **Igneous** and **metamorphic** rocks contain the highest quantities of magnetic minerals and have the strongest magnetic fields. **Sedimentary** rocks are weakly magnetic or non-magnetic because they contain negligible amounts of magnetic minerals (Australian Government, 2010).

- The remnant magnetism of a rock can be measured using a magnetometer. A **magnetometer** is any device capable of measuring the strength, direction or changes in Earth's magnetic field (Ungvarsky, 2023).

- Smart devices like smartphones and tablets contain built-in **3-axis magnetometer sensors** (NASA, 2025).

- When you bring a rock up to a magnetometer and take a reading, you are measuring Earth's ambient magnetic field strength (the background) plus the rock's own magnetic field. To isolate the rock's field, you must subtract the ambient field from the total field measurement (Halekas et al., 2010).

- A **magnetic anomaly** is a variation in the strength of Earth's magnetic field caused by magnetism in subsurface rocks (Luyendyk, 2025). Strongly magnetic rocks will create large magnetic anomalies.

- Geoscience Australia carry out **magnetic surveys** using magnetometers – these can be marine or airborne (Australian Government, 2010). These magnetic surveys quantify the magnetic anomalies within an area.

- Data from geomagnetic surveys is used to map the distribution of magnetic rocks, which can help identify rocks containing economically valuable mineral resources (Australian Government, 2010).

- Scientists do not generally quantify rock magnetism values in microtesla (μT). This is because there is no single μT value for a rock. The reading depends on:
  - The size and shape of the sample
  - The orientation of the sample
  - The distance between the sensor and the sample
  - Earth's background magnetic field

  The magnetism of rocks is generally reported in terms of **magnetic susceptibility.** This is described by the U.S. EPA (2016).

In this class, students compare the relative effect of different rock types on the magnetometer sensor in smart devices.

References:

Australian Government (2010). *Magnetics*. Available at: https://www.ga.gov.au/scientific-topics/disciplines/geophysics/magnetics

Halekas, J. S., R. J.Lillis, R. P.Lin, M.Manga, M. E.Purucker, and R. A.Carley (2010). How strong are lunar crustal magnetic fields at the surface?: Considerations from a reexamination of the electron reflectometry technique, *J. Geophys. Res.*, 115, E03006. https://doi.org/10.1029/2009JE003516

Luyendyk, B. P. (2025). Oceanic crust. *Encyclopedia Britannica*. Available at: https://www.britannica.com/science/oceanic-crust

NASA (2025). *Introductory Experiments in Magnetism with Smart Devices*. Available at: https://science.nasa.gov/learn/heat/resource/introductory-experiments-in-magnetism-with-smart-devices/

Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

University of British Columbia (2018). *Magnetic susceptibility*. Available at: https://www.eoas.ubc.ca/courses/eosc350/content/foundations/properties/magsuscept.htm

Ungvarsky, J. (2023). *Magnetometer*, EBSCO Research Starters. Available at: https://www.ebsco.com/research-starters/history/magnetometer

U.S. Environmental Protection Agency (EPA) (2016). *Magnetic methods*. Available at: https://archive.epa.gov/esd/archive-geophysics/web/html/magnetic_methods.html

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Additional resources:

For further information on crustal magnetism and how it is studied: Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

For additional classroom activities using smart device magnetometers:

NASA (2025). *Introductory Experiments in Magnetism with Smart Devices*. Available at: https://science.nasa.gov/learn/heat/resource/introductory-experiments-in-magnetism-with-smart-devices/

NASA (2025). *Exploratory Activities in Magnetism with Smart Devices*. Available at: https://science.nasa.gov/learn/heat/resource/exploratory-experiments-in-magnetism-with-smart-devices/

This class can be extended to involve the concepts of magnetic striping, seafloor spreading and plate tectonics (see **Figure 1**). The following resources are good starting points:

A YouTube video to introduce the concept of magnetic striping: Science with Thomas Stevenson (2019). *Magnetic Striping and Seafloor Spreading.* Available at: https://youtu.be/JJEZ3Vizdww?si=OMpPrh7v5m0CSF06

A simple hands-on classroom activity to model the process of seafloor spreading using paper strips, a cardboard box, a bar magnet and a compass:

Teacher instruction sheet: Teach Earth Science. *Modeling seafloor spreading: Teacher instruction sheet.* Available at: chrome https://teachearthscience.org/Resources/SeafloorSpreading%20_Teacher.pdf

Student data sheet: Teach Earth Science. *Modeling seafloor spreading: Student data sheet.* Available at: https://teachearthscience.org/Resources/SeafloorSpreading_Student.pdf

A more involved (undergraduate level) data analysis activity where students interpret marine magnetic anomalies and calculate seafloor spreading rates: Herrstrom, E. (2019). *Seafloor spreading: Bathymetry, anomalies, and sediments.* Cutting Edge. Available at: https://serc.carleton.edu/NAGTWorkshops/intro/activities/221148.html

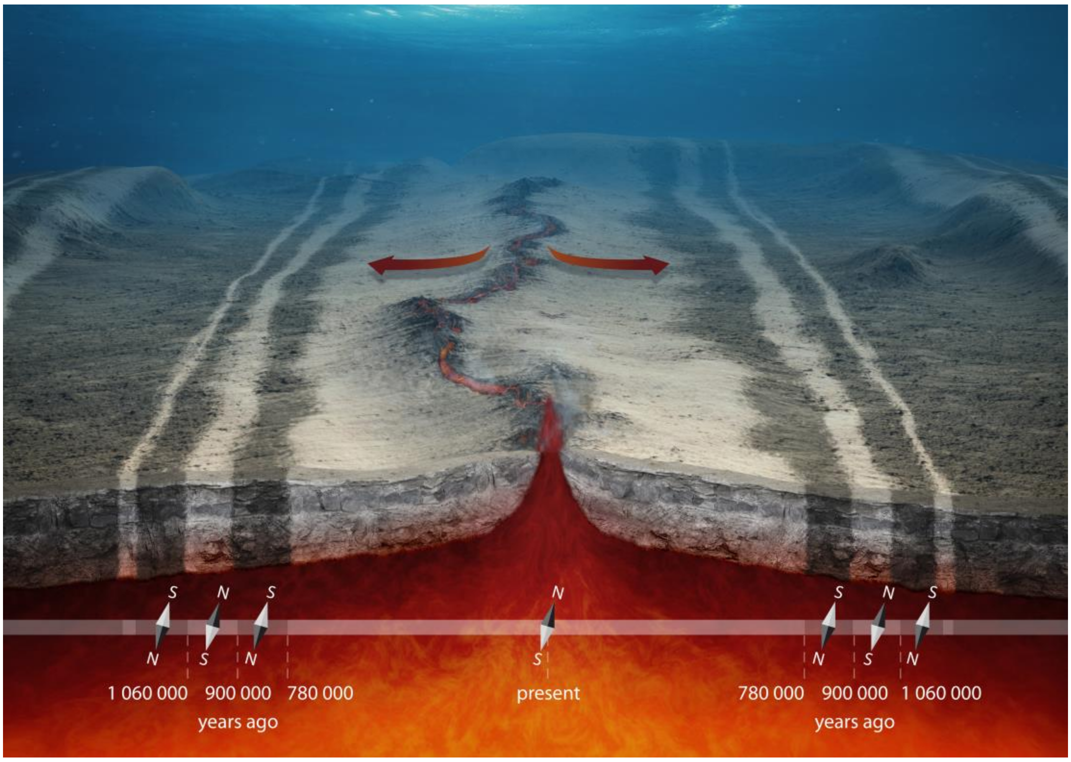

Figure 1: Magnetic striping on the seafloor at a mid-ocean ridge (ESA/AOES Medialab)